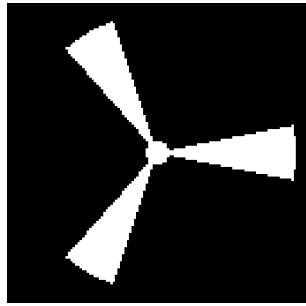
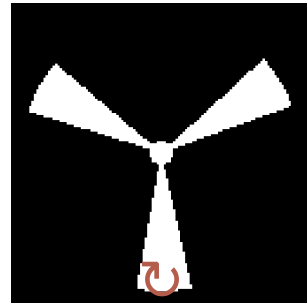


Supplementary Fig. S1 | Propeller controls validate decoded-flow direction

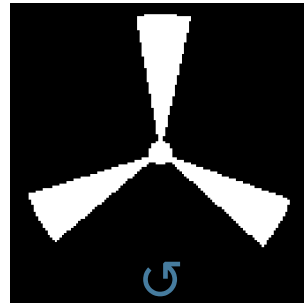
A Static (stationary)



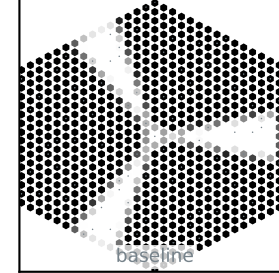
CW_mid (clockwise)



CCW_mid (counterclockwise)

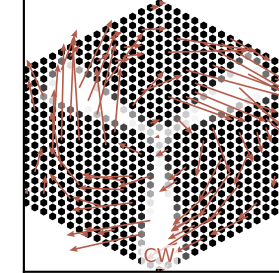


B Static

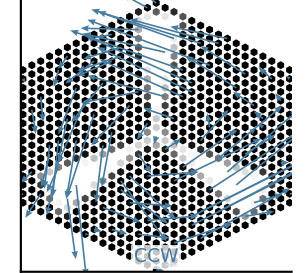


median-P representative: model 005 · frame 30

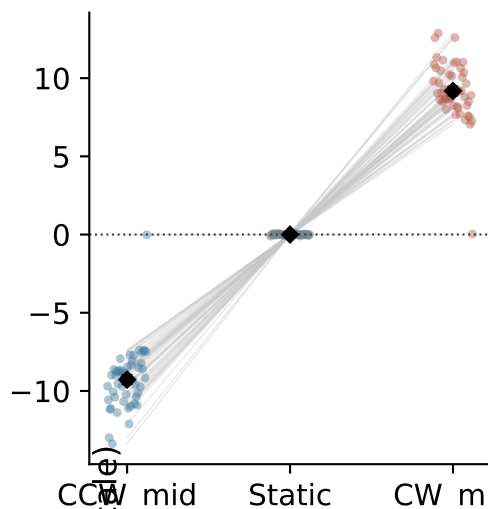
CW_mid



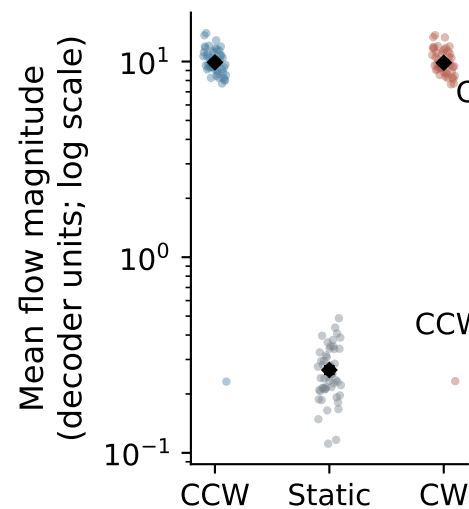
CCW_mid



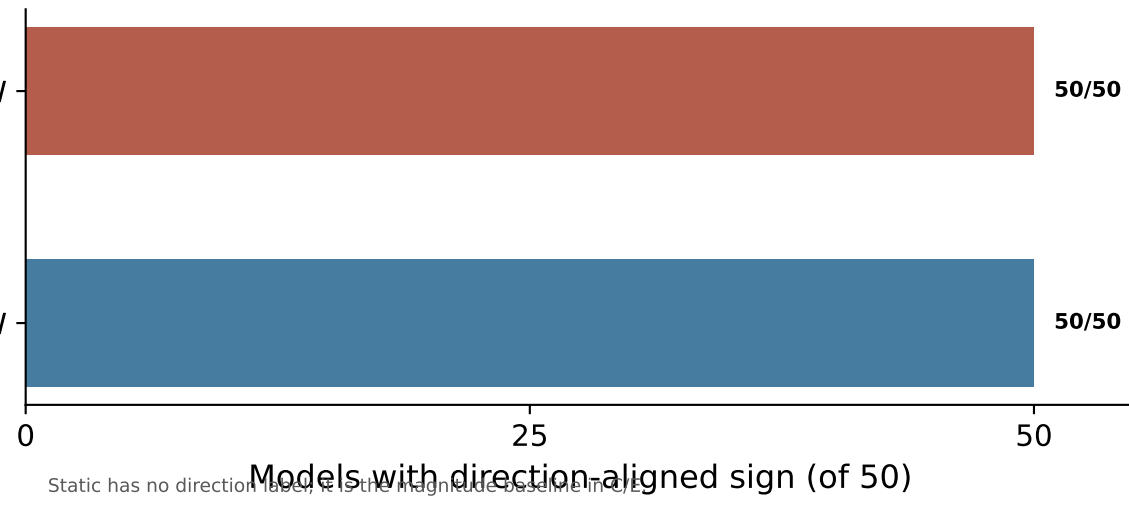
C Signed rotation metric · 50 models



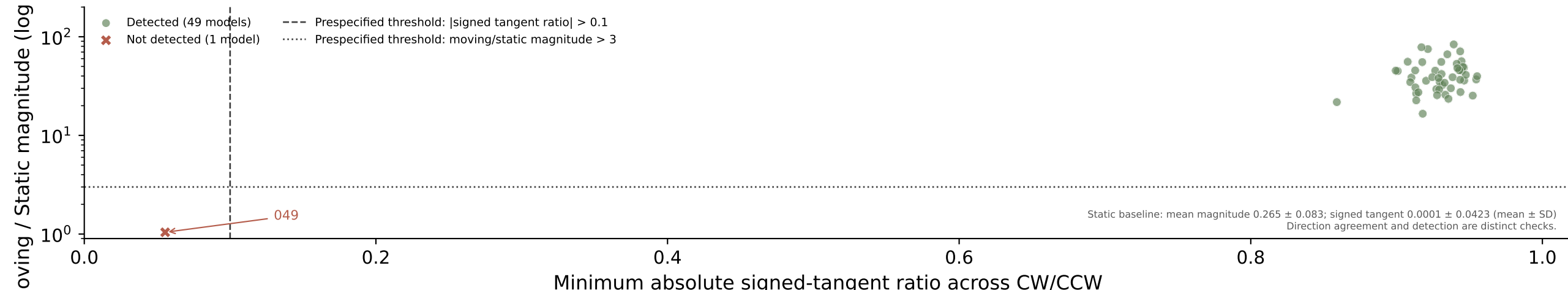
Magnitude (separate from sign)



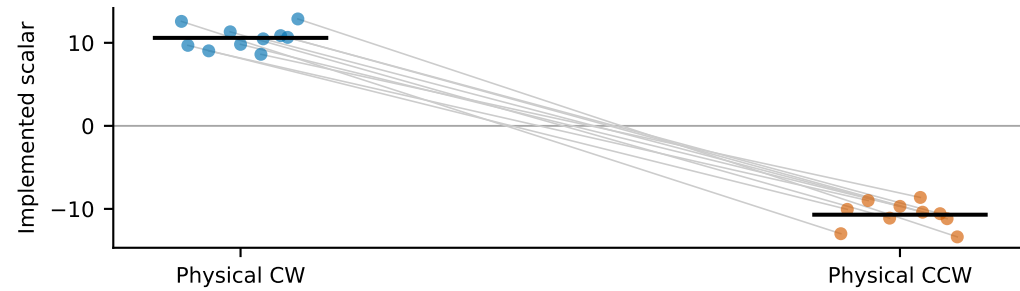
D Direction-label agreement



E Rotation detection uses both prespecified thresholds · 49/50 detected



Independent N10 calibration: single



Independent N10 calibration: multi

